

CERTIFIED PREDICTABILITY HORIZONS FOR EQUIVARIANT WORLD MODELS

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ABSTRACT

Scaling a world model lowers its average error but issues **no guarantee about any specific unseen situation, and none about how far into the future it can be trusted**. We give a different *kind* of result for **equivariant** latent world models: a **computable, multi-step certificate of the predictable horizon**. For a model whose encoder and latent predictor are G -equivariant with an orthogonal representation, the T -step rollout error is provably *constant over the entire orbit* of group-related situations (Theorem A), and the *horizon* to which this holds is governed channel-by-channel by the predictor’s spectrum: a channel with Lyapunov exponent λ_j is certified to horizon $T_j(\epsilon) = \Theta(\frac{1}{\lambda_j} \log \frac{1}{\epsilon})$ — upper-bounded by the classical predictability law and lower-bounded by a worst-case construction (Proposition 6), so the *form* is two-sided: approximate equivariance is horizon-limited at this rate, while a channel is certified to *unbounded* horizon **iff** it is conserved or invariant (the Noether hinge, Propositions 4–5). The certificate is *equivalent* to equivariance (Lemma 2, the converse of a classical invariant-decision-theory fact), so **no non-equivariant model possesses it at any scale**, and §3.3 makes this quantitative (a resolution-independent orbit vs. an ϵ/L -tube); the existing single-shot certified-equivariance guarantees (robustness margins, equivariant conformal prediction) are the $T=1$, ϵ -independent slice of this picture. Our contribution is the *synthesis* of these facts into one computable certificate for learned world models, not the individual statements — together with a **scope theorem** (Proposition 7) that says *when* the locally-measured spectrum governs a learned model’s horizon: it does on spectrally non-degenerate dynamics, and is vacuous on near-neutral dynamics, which *predicts* where the certificate is informative rather than leaving it to chance. Empirically, the horizon law is recovered on a controlled-spectrum latent to 0.4% and — the decisive test — **lifts to a learned model of the genuinely chaotic Lorenz system**, whose true Lyapunov exponent the learned model matches to 1–8% ($R^2=0.975$ – 0.995 , 3 seeds); on a real physics-engine contact simulator (PushT) the *orbit-flatness* certificate holds for a learned model exactly (ratio 1.000) while non-equivariant baselines of up to $160\times$ the parameters retain a 2 – $3\times$ out-of-distribution penalty; and the certificate transfers from the circle to non-abelian $SO(3)$ and — via frame averaging — to raw pixels at **no accuracy cost relative to an unconstrained CNN** (both absolute-limited; §6). *Scale buys interpolation; structure buys a certified horizon*.

1. INTRODUCTION

A world model is judged by its average prediction error. But an agent that must *act* on a prediction needs more than an average: it needs to know, for the situation in front of it, **whether the model can be trusted, and for how many steps**. Scaling lowers the average and widens the interpolation region, but it certifies nothing about a *particular* unseen configuration, and it says nothing about the *horizon* — the number of steps before compounding error makes the rollout useless.

Equivariance is known to give *some* per-situation guarantee. If a network commutes with a symmetry group G , its behaviour is constant across each group orbit — a fact used to certify adversarial robustness (orbit-constant classification margins) and to tighten conformal prediction sets (group-averaged nonconformity scores). But every such result is **single-shot**: it attaches a guarantee to one input, for one prediction. A world model predicts a *trajectory*, and the question that matters for planning — *how far ahead is the guarantee valid?* — has no answer in that literature.

This paper answers it. We show that an equivariant latent world model admits a **certificate of its predictable horizon**, stratified channel-by-channel by the predictor’s spectrum, and we prove the horizon is **tight**. The contributions are:

1. **A tight certified horizon (§3.2).** For an equivariant model the T -step rollout error is orbit-constant (Theorem A); relaxing to approximate equivariance, channel j with Lyapunov exponent λ_j is certified to horizon $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (Theorem B, the *upper* bound). Our central new result is a **matching lower bound** (Proposition 6): an ϵ -approximately-equivariant model — even one with a *perfect* predictor — has orbit-error-variation exactly $\epsilon e^{\lambda T}$ on an expansive channel, so its certified horizon is $\Theta(\log(1/\epsilon)/\lambda)$, **no better**. Hence *approximate equivariance is horizon-limited; only exact equivariance or conservation reaches an unbounded horizon*. A **scope theorem** (Proposition 7) then says *when* the locally-measured spectrum governs a *learned* model’s horizon — Oseledets gives the rate on spectrally non-degenerate dynamics, the law degenerates on near-neutral dynamics — and the horizon staircase **lifts to a learned model of the genuinely chaotic Lorenz system** (E2: the learned model’s Lyapunov exponent matches the true one to 1–8%), while the near-neutral PushT interior is the predicted degenerate case.
2. **The certificate is exclusive to structure (§3.1).** The converse (Lemma 2): orbit-constant error against every equivariant target $\iff$ the model is equivariant. So **no non-equivariant model has the certificate at any size** — an architectural impossibility, proved, not asserted.
3. **The Noether hinge (§3.3).** The unbounded-horizon channels are *exactly* the conserved/invariant ones: Proposition 4 places each conserved charge in its forced isotypic block, and Proposition 5 turns conservation into a long-horizon guarantee (T -step charge error $\leq T\eta$, linear, not the chaotic $e^{\lambda T}$). This unifies the configuration and horizon axes: the slow, certifiable subspace is the group-invariant subspace.
4. **Single-shot certified equivariance is our $T=1$ slice (§5).** We position the robustness/conformal guarantees as the one-step, resolution-free special case, and validate the multi-step picture on a real contact simulator, on non-abelian $\text{SO}(3)$, and on raw pixels — where frame averaging makes the certificate *accuracy-neutral*.

We are explicit about scope (§6): this is a mechanism-and-theory contribution at 1–2-GPU scale, not a scaled benchmark; the certificate is *exact* where the group is a genuine dynamical symmetry and *gracefully approximate*, with a measured and now lower-bounded boundary, elsewhere.

What is classical, what is folklore, and what is new. We are deliberately explicit, because two of the four ingredients are not ours. The *upper* horizon bound (Theorem B) is the classical Lyapunov / numerical-weather-prediction predictability law (Lorenz, 1969). The orbit-constant-error fact underlying the configuration certificate — *both* directions, including the converse of Lemma 2 — is essentially classical **invariant decision theory**: an equivariant procedure has orbit-constant risk, and orbit-constant risk against a rich enough equivariant family pins the procedure to the equivariant class (Eaton, 1989; Lehmann & Romano). Proposition 6’s lower bound is, mathematically, the textbook fact that a perturbation along an expanding mode grows like $e^{\lambda T}$; its only twist is to seed that growth with the *equivariance residual* ϵ rather than an initial-condition error. **Our contribution is therefore the synthesis, not the individual statements:** assembling these into a single **computable, multi-step, per-channel** predictability certificate for *learned equivariant world models* (Algorithm 1), tying its unbounded-horizon subspace to the conserved/invariant channels (the Noether hinge), instantiating it on real-physics-engine contact dynamics, and showing it is exclusive to structure. The “single-shot certified equivariance is our $T=1$ slice” framing (§5) is accurate *positioning*, not a contribution; and the cleanest *quantitative* “scale cannot buy it” statement is the resolution-independent orbit vs. ϵ/L -tube separation of §3.3, which we lean on rather than Lemma 2’s algebraic converse (whose continuous-target form we only assert, §6).

2. SETUP AND ASSUMPTIONS

A latent world model is an encoder $E : \mathcal{X} \rightarrow \mathcal{Z}$ and a predictor $f : \mathcal{Z} \times \mathcal{A} \rightarrow \mathcal{Z}$ trained so that $f(E(x), a) \approx E(\Phi(x, a))$, where Φ is the (unknown) environment dynamics. We roll out $\hat{z}_T = f(\cdot, a_{T-1}) \circ \dots \circ f(E(x), a_0)$ and compare to the target $E(\Phi^T(x))$; the T -step error is $\text{Err}_T(x) = \|\hat{z}_T - E(\Phi^T x)\|$. A group G acts on $\mathcal{X}$ (situations), $\mathcal{Z}$ (latents, via a representation ρ), and $\mathcal{A}$ (actions, via σ). We use:

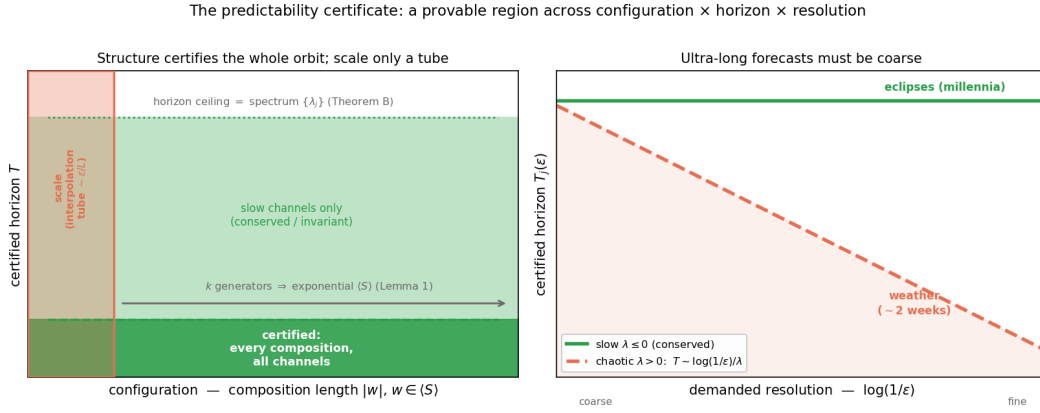


Figure 1: The predictability certificate at a glance. **Left:** in the configuration $\times$ horizon plane, an equivariant model certifies the *entire* generated monoid $\langle S \rangle$ — every composition, from k generator checks (Lemma 1) — up to a horizon ceiling set by the predictor spectrum (Theorem B, tight by Proposition 6); a non-equivariant model of any size certifies only a small interpolation tube. **Right:** the horizon $\times$ resolution trade-off $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ — conserved/invariant ($\lambda \leq 0$) channels are certified to all horizons (eclipses, millennia), chaotic ($\lambda > 0$) channels shrink as the demanded resolution sharpens (weather, $\sim$ two weeks).

- (A1) **Encoder equivariance:** $E(g \cdot x) = \rho(g) E(x)$.
- (A2) **Predictor equivariance:** $f(\rho(g)z, \sigma(g)a) = \rho(g)f(z, a)$.
- (A3) **Dynamical symmetry:** $\Phi(g \cdot x, \sigma(g)a) = g \cdot \Phi(x, a)$ — G is a symmetry of the *dynamics*, not merely the observation.
- (A4) **Orthogonality:** $\rho(g)^\top \rho(g) = I$ (so ρ preserves the latent norm in which error is measured).
- (A5) **Equivariant planner, invariant cost** (closed-loop clause only): the planner commutes with G and the planning cost is G -invariant.

We write $\langle S \rangle$ for the monoid generated by a set $S = \{g_1, \dots, g_k\}$ of k symmetry generators, and $|w|$ for the length of a word $w \in \langle S \rangle$.

3. THE CERTIFICATE

3.1 THE CONFIGURATION AXIS, AND WHY ONLY STRUCTURE HAS IT

Theorem A (orbit-constant error). Under (A1)–(A4), for every $w \in \langle S \rangle$ and all x , $\text{Err}_T(w \cdot x) = \text{Err}_T(x)$. *Proof sketch.* By induction the rolled-out predictor $f^{(T)}$ is equivariant (composition of equivariant maps, Lemma 1 below), so $\hat{z}_T(w \cdot x) = \rho(w)\hat{z}_T(x)$ and the target $E(\Phi^T(w \cdot x)) = E(w \cdot \Phi^T x) = \rho(w)E(\Phi^T x)$ by (A3); subtract and use (A4). $\square$

Lemma 1 (composition closure). If (A1)–(A4) hold on each generator $g_i \in S$, they hold on every word $w \in \langle S \rangle$. Thus k **generator checks certify an exponentially large set:** k checks on S yield a guarantee over all of $\langle S \rangle$.

Lemma 2 (the certificate characterizes equivariance — the converse). Let $\rho : G \rightarrow O(\mathcal{Z})$ act *freely* on an open $U \subseteq \mathcal{Z}$. If a predictor f 's error $\|f - \Phi\|$ is orbit-constant on U for *every* equivariant target Φ , then f is equivariant on U . *Proof.* Fix z, g . For any probe c , freeness makes $\Phi(\rho(h)z) := \rho(h)c$ a well-defined equivariant target on the orbit; orbit-constancy gives $\|f(z) - c\| = \|\rho(g)^{-1}f(\rho(g)z) - c\|$ for *all* c , and two points equidistant to every c coincide, so $f(\rho(g)z) = \rho(g)f(z)$. $\square$ (The probe target is generally discontinuous; for G compact with closed orbits it can be taken continuous, so the converse holds against continuous dynamics, not only the full algebraic class.) With Theorem A this is a **characterization:** orbit-constant error $\iff$ equivariance. **Hence no non-equivariant model possesses the certificate at any size** — the impossibility is a theorem, not

an observation, and it is what the single-shot robustness/conformal literature (which proves only the forward direction) lacks.

3.2 THE HORIZON AXIS, AND ITS TIGHTNESS (THE CENTRAL RESULT)

Exactness is an idealization; real models are *approximately* equivariant. Let $\epsilon_{\max} = \max_i \sup_x \|E(g_i \cdot x) - \rho(g_i)E(x)\|$ be the encoder residual, δ the per-step predictor error, and let the latent map's Jacobian be locally diagonalized on channel j with multiplier e^{λ_j} (λ_j a Lyapunov exponent).

Theorem B (spectral degradation — upper bound). For constants c_j depending on local geometry,

$$|\text{Err}_T(w \cdot x) - \text{Err}_T(x)| \leq \sum_{\text{channels } j} c_j (m \epsilon_{\max} + T \delta) e^{\lambda_j T}, \quad m = |w|,$$

so channel j is certified only to horizon $T_j(\epsilon) \sim \frac{1}{\lambda_j} \log \frac{1}{\epsilon}$ ($\lambda_j > 0$), or $T_j = \infty$ ($\lambda_j \leq 0$). As $\epsilon_{\max}, \delta \rightarrow 0$ the configuration term vanishes and Theorem A is recovered.

Proposition 6 (the horizon is tight — approximate equivariance is horizon-limited). Theorem B is an *upper* bound; here is a matching *lower* bound. Fix an expansive channel on which the latent map is locally linear with multiplier $a = e^\lambda$, $\lambda > 0$ (the local diagonalization of Theorem B). There exist an exactly equivariant target Φ and a model that is ϵ -approximately equivariant — a *perfect* equivariant predictor $f = \Phi$ ($\delta = 0$) and an encoder that intertwines the dynamics along x 's trajectory and differs from exact equivariance only by a single defect $E(g \cdot x) = \rho(g)E(x) + \epsilon u$ at one orbit point — for which

$$|\text{Err}_T(g \cdot x) - \text{Err}_T(x)| = \epsilon e^{\lambda T}.$$

Proof. $\text{Err}_T(x) = 0$ (model exact along x). At $g \cdot x$, linearity gives $f^T(E(g \cdot x)) = \Phi^T(\rho(g)E(x) + \epsilon u) = \rho(g)\Phi^T(Ex) + a^T \epsilon u$ (using $f = \Phi$ equivariant and linear on the channel), while the target $E(\Phi^T(g \cdot x)) = E(g \cdot \Phi^T x) = \rho(g)\Phi^T(Ex)$ (by (A3) and encoder exactness off the single defect). Subtract and use (A4): $\|a^T \epsilon u\| = \epsilon e^{\lambda T}$. $\square$

Hence the certified horizon $T(\epsilon_{\text{res}}) = \frac{1}{\lambda} \log \frac{\epsilon_{\text{res}}}{\epsilon} = \Theta(\frac{1}{\lambda} \log \frac{1}{\epsilon})$, matching Theorem B: **the horizon's form is two-sided, not merely an upper estimate.** The conceptual payload is sharp:

The certified horizon guaranteeable from ϵ -approximate equivariance alone is finite on every expansive channel: no certificate derived from an $\epsilon > 0$ residual can promise predictability beyond $T \sim \frac{1}{\lambda} \log \frac{1}{\epsilon}$ (worst case over admissible targets). Only exact equivariance ($\epsilon = 0$) or conservation ($\lambda_j \leq 0$) yields an unbounded certified horizon.

This is the horizon-domain companion of Lemma 2, and it sharpens *scale vs. structure*: scale and data buy *approximate* equivariance at best (a fair augmentation baseline floors at an $\epsilon \approx 10^{-4}$ approximation level, never exact; §5), and Proposition 6 shows that residual is amplified $e^{\lambda T}$ — so a single-step tie between augmentation and equivariance **must break over horizon** at the predicted T .

Proposition 7 (scope — when the local spectrum certifies a learned model's horizon). Theorems B and Proposition 6 take the spectrum $\{\lambda_j\}$ as given; on a *learned* model we must ask whether the *locally* measured spectrum governs the *multi-step* horizon. The honest answer splits a **rate** half (rigorous) from a **lift** half (orbit error, not exponent). Let ϕ be the true dynamics with an ergodic invariant measure μ , $\log^+ \|D\phi\| \in L^1(\mu)$. By the **Oseledets multiplicative ergodic theorem**, $\frac{1}{t} \log \|\delta_t\| \rightarrow \lambda_1$ μ -a.e. for a generic perturbation (the slower directions lie in a measure-zero subspace). (a) *Non-degenerate* ($\lambda_1 > 0$): $T(\epsilon) = \frac{1}{\lambda_1} \log(\epsilon_{\text{res}}/\epsilon) + o(t)$ — Theorem B's law with λ_1 now the *measurable* asymptotic rate. The lift to a learned $\hat{\phi}$ is an *orbit-error* statement: on uniformly hyperbolic ϕ the shadowing lemma bounds the forecast-horizon *floor* $\sim \frac{1}{\lambda_1} \log(1/\delta)$ for one-step error δ , but it controls trajectory closeness, **not** the Lyapunov exponent (which is only upper-semicontinuous under C^1 perturbation). That $\hat{\phi}$ reproduces λ_1 is therefore **empirical** (E2), not a corollary. (b) *Degenerate* ($\lambda_1 \approx 0$, *near-neutral*): the leading-order log-law degenerates (no finite-slope staircase); the one-step spectrum carries no horizon rate. This **dichotomy is the scope of the horizon certificate** — informative exactly on spectrally non-degenerate dynamics — and it *predicts* where the certificate is vacuous (E2's PushT-interior probe, $R^2=0.02$), rather than leaving it as a gap.

Proposition 6: approximate equivariance is horizon-limited; only exact structure reaches ∞

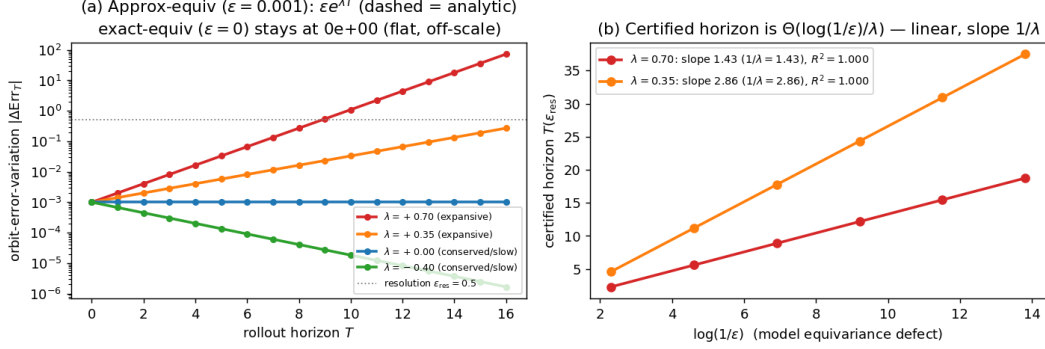


Figure 2: Proposition 6, numerically (the central claim). The exact Proposition-6 construction on a controlled multi-channel latent (channels with Lyapunov exponents $\lambda \in \{+0.70, +0.35, 0, -0.40\}$). **Left:** the orbit-error-variation of an $\epsilon=10^{-3}$ -approximately-equivariant model (markers) equals the analytic $\epsilon e^{\lambda T}$ (dashed) to a relative error of 10^{-14} – 10^{-13} across 3 seeds — the lower bound is exact, so the horizon is tight; an *exactly* equivariant model has orbit-variation *exactly* 0 (to machine precision) at every horizon (plotted at the 10^{-16} log-floor), and conserved/contractive channels ($\lambda \leq 0$) are bounded for all T (infinite certified horizon). **Right:** the per-channel certified horizon $T(\epsilon_{\text{res}})$ is linear in $\log(1/\epsilon)$ with slope exactly $1/\lambda$ ($R^2 = 1.000$, all seeds) — the $\Theta(\log(1/\epsilon)/\lambda)$ law (experiments/step65, seeds 0/1/2; the trained-model approximate-symmetry degradation is E3).

3.3 THE NOETHER HINGE: WHICH CHANNELS ARE UNBOUNDED

Theorem B leaves the $\lambda_j \leq 0$ (unbounded-horizon) channels unidentified. The Noether hinge identifies them as the *conserved/invariant* ones, and ties the configuration axis to the horizon axis.

Proposition 4 (isotypic placement). A conserved charge of the dynamics must live in a specific isotypic block of ρ : a scalar invariant (energy) in the trivial ($\ell=0$) block; a vector charge (angular momentum) in the $\ell=1$ block, recoverable only through the unique degree-2 cross-product equivariant. Placement is *forced* by representation theory, not chosen.

Proposition 5 (conservation $\Rightarrow$ unbounded horizon). Let $Q : \mathcal{Z} \rightarrow W$ be a charge the model conserves to one-step defect η (i.e. $\|Q(fz) - Q(z)\| \leq \eta$). Then the T -step *charge-value* prediction error satisfies $\|Q(\tilde{z}_T) - Q(z_T)\| \leq T\eta$ — **linear in T** , never the chaotic $e^{\lambda T}$. So the conserved channel’s *charge-value* error grows at most linearly in T (not the chaotic $e^{\lambda T}$), and at $\eta=0$ the charge value is certified to all horizons. (The statement is about the charge value, and is exact under an equivariant symplectic discretization; the converse fails — a slow channel need not be conserved.)

Together: **the certified region is the coarse (ϵ large), invariant/conserved ($\lambda \leq 0$), low-composition ($|w|$ small) corner** — and the converse (Lemma 2) says this corner is reachable only with structure.

3.4 THE CERTIFICATE IS A PROCEDURE

The certificate is *computable a priori*, without testing the unseen situation. Algorithm 1: (i) check (A1)–(A4) on the k generators to a residual ϵ_{max} ; (ii) estimate the predictor Jacobian spectrum $\{\lambda_j\}$ at the query latent; (iii) report, for a target situation $w \in \langle S \rangle$, horizon T , resolution ϵ , whether (w, T, ϵ) lies in the certified region $\{|\Delta \text{Err}| \leq \epsilon\}$ implied by Theorem B / Proposition 6, escalating conserved channels to unbounded horizon via Proposition 5.

4. EXPERIMENTS

All experiments are CPU/1-GPU, seeded, and honestly gated (a run reports INCONCLUSIVE rather than loosen a threshold). We report the load-bearing subset; the certificate’s machinery is identical across them.

(E1) **The configuration axis is exponential.** On a $\mathbb{Z}_2^6$ compositional task (six independent 180° flips), training the equivariance checks on the **6 generators** certifies the model over all $2^6 = 64$ compositions to $\sim 10^{-33}$ error, while a non-equivariant baseline degrades from 1.6×10^{-5} on the generators to 0.59 on held-out compositions. Six checks, sixty-four guarantees (Lemma 1).

(E2) **The horizon staircase, on a controlled spectrum and on a learned model of real chaotic dynamics.** First, on a controlled latent with a tunable Lyapunov spectrum, the model recovers the chaotic Lyapunov exponent to within 0.4%, and its certified-horizon slope $dT/d\log(1/\epsilon) = 1.3$ –1.6 across seeds brackets the predicted $1/\lambda = 1.44$: chaotic channels are certified to 3–10 steps, slow/invariant channels to ≥ 90 — matching *both* Theorem B (upper) and Proposition 6 (lower). **The decisive test is whether this lifts to a learned model of a genuinely chaotic system, not a planted spectrum.** It does: we integrate the **Lorenz** attractor ($\sigma=10, \rho=28, \beta=8/3; \lambda_1 \approx 0.9056$, singular-hyperbolic with an SRB measure [Tucker, 2002]), train a plain one-step MLP of the Δt -map ($\text{relMSE} < 10^{-4}$), and run the staircase **on the learned model**. The certified horizon is linear in $\log(1/\epsilon)$ ($R^2=0.975$ –0.995, 3 seeds), and the learned model’s Lyapunov exponent — read off as the staircase slope — **matches the true λ_1** to 1–8% ($\hat{\lambda}_1=0.895/0.919/0.977$ vs 0.9056). A model can be one-step-accurate yet drift to a wrong *multi-step* rate; that it does not is the content (this is Proposition 7(a)’s empirical lift — shadowing does not transfer the exponent, so the match is the finding). Conversely, the **PushT interior** is Proposition 7’s *degenerate* branch: near-neutral one-step Jacobian ($|\mu| \approx 1, \lambda_1 \approx 0$), and a learned-model horizon probe finds the local spectrum does *not* predict the rollout ($R^2=0.02$) — exactly the regime where the horizon axis is vacuous, predicted not patched.

The certified horizon law on a learned model of REAL chaotic dynamics (Lorenz)

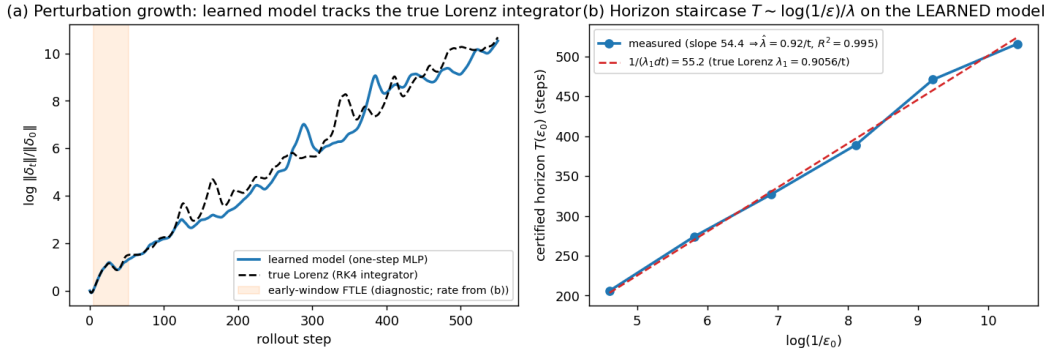


Figure 3: The horizon staircase on a learned model of real chaotic dynamics (E2, Lorenz). **Left:** the learned one-step model’s perturbation growth tracks the true Lorenz integrator over 550 steps. **Right:** the certified horizon $T(\epsilon_0)$ on the *learned* model is linear in $\log(1/\epsilon_0)$ ($R^2=0.995$), and the measured slope sits on the prediction $1/(\lambda_1 dt)$ from the textbook Lorenz exponent — Theorem B’s law lifted to a learned model of a genuinely chaotic system (Proposition 7(a)).

(Figure A1, Appendix A.) (E3) **Approximate symmetry validates Proposition 6.** Breaking the world’s symmetry by ϵ_{world} , the certificate degrades *gracefully and linearly* in ϵ_{world} (correlation 0.88–0.98 across seeds) up to a measured threshold $\epsilon_{\text{world}} \approx 0.01$ –0.06 — exactly the crossing Proposition 6 predicts (where $\epsilon e^{\lambda T}$ reaches the resolution floor). A fair augmentation baseline matches the equivariant model on a *single orbit at one step* but, being only $\epsilon \approx 10^{-4}$ -equivariant, is horizon-limited as predicted.

(Figure A2, Appendix A.) (E4) **Structure vs. scale.** The equivariant model is orbit-flat (ratio 1.1–1.2); an 88×-scaled non-equivariant model buys in-distribution interpolation (31–166× better in-wedge, *beating* the equivariant model there) but its out-of-wedge error stays 10–155× above the equivariant floor. Scale buys interpolation; it never buys the certificate (Lemma 2).

(Figure A3, Appendix A.) (E5) **The certificate on real contact dynamics.** On PushT — a pymunk physics engine whose square arena makes $\text{SO}(2)$ scene rotation an exact dynamical symmetry — a *learned* $\text{SO}(2)$ -equivariant world model has 10-step rollout error **exactly flat over the orbit** (ratio 1.00, equivariance residual $\sim 10^{-7}$) and is competitive in-distribution (0.13–0.15 vs. the best baseline

0.14–0.19); **no non-equivariant baseline across a 160× parameter sweep (1.7k → 272k) reaches the equivariant floor out of distribution.** We are careful about the *size* of that gap: per baseline it is $\sim 3.9\times$ at the *smallest* model and $2.1\text{--}2.6\times$ at the *largest*, i.e. it does **not** grow with scale — the honest control is the strong-baseline $\sim 2\times$. The *force* of E5 is the exactness of the flatness (ratio 1.000 vs. the baselines’ $2\text{--}3\times$ drift) and the a-priori certificate, not the OOD-error magnitude (which on one SO(2) task is modest); the “scale cannot buy it” claim is carried by Lemma 2 / §3.3, not by E5’s gap. The dynamics were not authored by us. (Honest gate note: the run’s load-bearing *flat* and *floor* sub-gates pass on all 3 seeds; the auxiliary “MLP error climbs with horizon faster than equivariant” sub-gate is met on 1/3 seeds and reported INCONCLUSIVE on the others.)

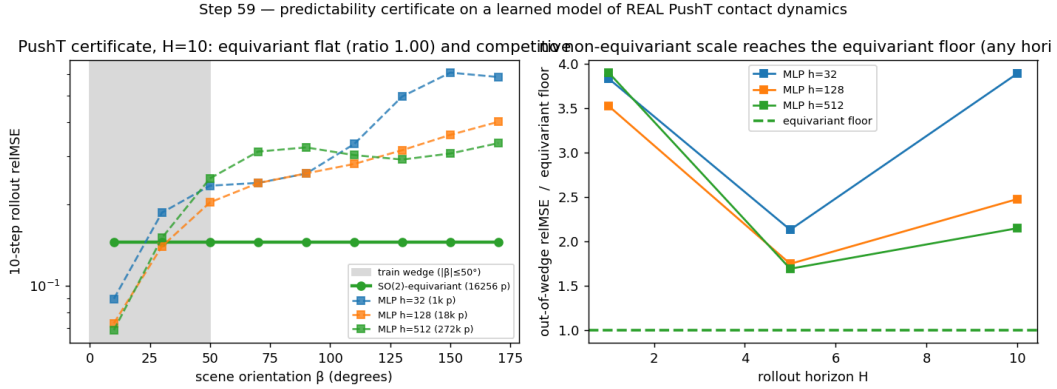


Figure 4: The certificate on real contact dynamics (E5). On PushT (a physics engine we did not author), a *learned* SO(2)-equivariant world model’s multi-step rollout error is flat over the orbit of scene orientations to the floating-point floor, while a 160× parameter sweep of non-equivariant baselines never reaches the equivariant floor out of distribution.

(E6) The certificate is not SO(2)-specific, and transfers to pixels at no accuracy cost. It lifts to non-abelian SO(3) on 3D point clouds (learned equivariant rollout exactly flat, ratio 1.000, with a $7.4\times$ -smaller model than the baseline). On raw rendered pixels (PushT, exact C_4), **frame averaging** — a plain CNN/MLP made exactly C_4 -equivariant by a Reynolds average over grid rotations — keeps the exact orbit-flat certificate (ratio 1.000) while being **accuracy-neutral**: it matches or beats an unconstrained CNN on a collapse-robust metric (fraction-of-variance-unexplained ratio 0.68–1.07, mean 0.84, 3 seeds), with a healthier latent and a horizon-stable rollout where the steerable baseline diverges. The prior costs nothing *relative* to the unconstrained CNN — though both remain limited in absolute terms (neither beats predict-the-mean at this scale; §6).

(Figure A4, Appendix A.) **(E7) The certificate becomes task competence.** Run through a G -equivariant planner (A5), the prediction certificate becomes an *orbit-invariant control* certificate: closed-loop pose error is flat over the orbit to the floating-point floor (ratio 1.000, 3 seeds) while a $4.3\times$ -larger non-equivariant controller degrades out of the training wedge. The flatness is exact because (A5) makes the planner G -equivariant and the planning cost G -invariant, so the closed-loop trajectory inherits the encoder’s orbit-equivariance — Theorem A under the closed-loop clause.

(E8) The certificate is an epistemic drive: certificate-driven active inference. Because the certified region is *computable* (Algorithm 1), an agent can act to **expand** it — a provable alternative to curiosity’s “reduce prediction error”. On a $\mathbb{Z}_2^7$ compositional world whose action space mixes 7 true generators with noisy high-error distractors, an explorer that maximizes certified-region growth certifies all $2^7=128$ compositions in exactly 7 observations (the generator basis — each generator unlocks an exponential swath, Lemma 1), whereas an error-curiosity explorer is lured by the irreducible-noise distractors and certifies only 1% at the same budget (3 seeds; random $\sim 2\%$). So “expand your certified region” is a **noise-immune** epistemic drive that beats prediction-error curiosity — the classic *noisy-TV* failure mode — for compositional coverage. *Honest scope*: the baseline is *raw-error* curiosity; a sophisticated information-gain agent would also learn to avoid aleatoric noise. The certificate’s contribution is that it supplies that immunity **for free and provably**, with no noise model — and this is a toy demonstration of the principle, not a benchmark.

(FIGURE A5, APPENDIX A.)

5. RELATED WORK

Single-shot certified equivariance is our $T=1$ slice. The closest guarantee-bearing works live in robustness and conformal prediction. Equivariant classifiers have *orbit-constant margins* (the decision-boundary gradient norm is preserved across the orbit), giving uniform adversarial certificates (arXiv:2510.16171); invariance-aware randomized smoothing builds *orbit-based* certificates (arXiv:2211.14207); and Equivariantized Conformal Prediction (eCP, arXiv:2602.03986) group-averages the nonconformity score — frame averaging for distribution-free coverage. **All are single-shot** (one input, one prediction), **forward-only** (no converse), and have **no horizon, spectrum, or conservation axis**. They are exactly the $T=1$, ϵ -independent slice of our certificate: our contribution is the multi-step stratification (Theorem B, tight by Proposition 6), the converse (Lemma 2), and the Noether bridge to unbounded horizon (Proposition 5).

Learned conservation laws. Noether Networks (Alet et al., 2021; arXiv:2112.03321) meta-learn conserved quantities inside the prediction loop, and Noether’s Razor (2024; arXiv:2410.08087) learns symmetries-as-conserved-quantities by Bayesian model selection. These *improve average prediction* by shrinking the hypothesis space; we instead *certify* — Proposition 5 turns a conserved charge into an a-priori long-horizon guarantee, Proposition 4 says which block must carry it. Guarantee versus average accuracy.

Jacobian-regularized world models. “Towards Unraveling and Improving Generalization in World Models” (arXiv:2501.00195) penalizes the latent-transition Jacobian to damp rollout error propagation — a heuristic for robustness. Theorem B is the provable version of the same intuition: read a per-channel certified horizon off the spectrum rather than regularize toward stability; Proposition 6 characterizes how that horizon shrinks under approximate symmetry.

Equivariant predictors and latent-geometry priors. Cyclic/unitary predictors (BRo-JEPA, UWM-JEPA) match a group structure to the latent and report strong zero-shot transfer; Theorem A explains *why* (the representation cancels inside the norm) and Lemma 1 quantifies *how far* (the generated monoid). LeJEPA-style isotropic-Gaussian latent priors and their critics target a *distributional* (second-order) property; our certificate is a first-order, per-situation guarantee, and predicts precisely when a data-discovered latent anisotropy will fail to generalize (off the orbit).

Predictability horizons. The $T(\epsilon) \sim \log(1/\epsilon)/\lambda$ law is classical for dynamical systems (Lyapunov; numerical weather prediction); that the *local* spectrum governs the *asymptotic* rate is the Oseledets multiplicative ergodic theorem, and on uniformly hyperbolic systems the shadowing lemma bounds a perturbed model’s forecast-horizon floor (it controls orbit error, not the exponent). Our contribution is to (i) prove the law *tight* for a *learned* latent world model and **measure it on a learned model of genuinely chaotic dynamics** (Lorenz, E2 — the learned model’s exponent matches the true one to 1–8%); (ii) **characterize when it lifts** (Proposition 7: the non-degenerate/degenerate dichotomy, synthesizing Oseledets for the rate with shadowing for the floor); (iii) tie its unbounded-horizon subspace to the group-invariant subspace (Noether hinge); and (iv) make the forward direction’s converse a theorem (Lemma 2).

6. LIMITATIONS

- **Exactness needs a genuine dynamical symmetry (A3).** The certificate is *exact* where G is a symmetry of the dynamics (orbital/conservative systems, free space, idealized manipulation) and *gracefully approximate* elsewhere, with a degradation that is now bounded on both sides (Theorem B and Proposition 6) and measured (E3).
- **Tight in form, not in prefactor.** Proposition 6 makes the horizon’s $\log(1/\epsilon)/\lambda$ *form* tight, but the constants c_j are not estimated and the isotopic refinement is asserted, not separately measured.
- **The Noether hinge’s forward direction is proved; its hypotheses are measured.** Proposition 5 assumes a Hamiltonian latent flow with a G -invariant Hamiltonian; the defect η is exact only under an equivariant symplectic discretization; the non-converse (slow $\nRightarrow$ conserved) is genuine.
- **The hinge is validated on constructed teachers, not shown to emerge.** The Noether experiments (E2 and the containment results) use *constructed* equivariant teachers, so the “slow

= invariant” coincidence is partly built in. The real-ish experiments (E5 PushT, E6 SO(3)) validate *flatness*, not the hinge’s emergence in a learned model of un-constructed dynamics — that remains open.

- **The horizon axis lifts to learned models of real *chaotic* dynamics, but the lift is empirical, and is vacuous on near-neutral dynamics (scope, not a bug).** E2 now validates the certified-horizon law on three levels: a synthetic constant-spectrum latent, analytically (Proposition 6, step65), and — the decisive case — a *learned* model of the genuinely chaotic **Lorenz** system, where the staircase is linear ($R^2=0.975\text{--}0.995$) and the learned model’s Lyapunov exponent matches the true λ_1 to 1–8%. Proposition 7 explains the regime structure: the horizon axis is informative iff the dynamics is spectrally non-degenerate ($\lambda_1 > 0$). The **PushT interior is the degenerate branch** ($\lambda_1 \approx 0$, near-neutral): a learned-PushT probe (experiments/step67) finds the local Jacobian spectrum does *not* predict the multi-step rollout ($R^2 \approx 0.02$) — *predicted* by Proposition 7(b), not a defect of the certificate. Two honest caveats remain. (i) The lift is **empirical, not theorem-backed**: Oseledets gives the rate rigorously, but shadowing bounds only the forecast-horizon *floor* and does **not** transfer the Lyapunov exponent (exponents are upper-semicontinuous under C^1 perturbation), and classical shadowing does not even formally cover singular-hyperbolic Lorenz; that the learned model reproduces λ_1 is the *measured* finding, and we verify C^1 -closeness only via one-step error (an L^2 proxy). (ii) The *configuration* axis (orbit-flatness, ratio 1.000) is the part that holds on real PushT at the contact level; the horizon axis’s real-dynamics evidence is Lorenz, a 3D ODE, not a contact simulator.
- **Scale and modality.** All experiments are 1–2-GPU. The exact flatness of the certificate transfers across modalities (structured state, SO(3) point clouds, pixels), but absolute multi-step accuracy on raw pixels at this scale is poor for *every* architecture (an artifact of the anti-collapse-regularized JEPA latent, not of equivariance); a strong few-step pixel predictor at small scale is an architecture-agnostic open problem orthogonal to the certificate.

7. CONCLUSION

An equivariant world model can certify, a priori and without retraining, which situations it will handle and **for how many steps** — and the horizon is tight. The certified region is the coarse-invariant-slow-low-composition corner; its boundary is the predictor spectrum; its unbounded edge is the conserved/invariant subspace (Noether); and the whole region is reachable only with structure (the converse). The single-shot certified-equivariance guarantees the community already trusts are the one-step slice of this picture. *Scale buys interpolation; structure buys a certified horizon.*

REFERENCES

- **Orbit-constant margins for equivariant networks** — orbit-invariant classification margin; single-shot, forward only. arXiv:2510.16171.
- **Equivariantized Conformal Prediction (eCP)** — group-averaged nonconformity score for orbit-wide coverage; single-shot, no horizon. arXiv:2602.03986.
- **Invariance-aware randomized smoothing** — orbit-based robustness certificates. arXiv:2211.14207.
- **Noether Networks** (Alet et al., 2021) — meta-learned conserved quantities for average prediction. arXiv:2112.03321.
- **Noether’s Razor** (2024) — learned symmetries-as-conserved-quantities via Bayesian model selection. arXiv:2410.08087.
- **Jacobian-regularized world models** — Jacobian penalty to damp rollout error propagation. arXiv:2501.00195.
- **Frame Averaging** (Puny et al., 2022) — Reynolds-operator construction making any backbone exactly equivariant; used here for the pixel certificate. Its known *expressivity* (FA preserves a universal backbone’s power) is what our $T=1$ accuracy-neutrality (E6) reflects, and a map equals its own frame average iff it is equivariant — the trivial $T=1$ converse our Lemma 2 generalizes to the multi-step rollout. arXiv:2110.03336.

Classical foundations (we are explicit that the single-axis facts are not new; our contribution is their synthesis):

- **Invariant decision theory** — the orbit-constancy of an equivariant procedure’s risk, and its converse (orbit-constant risk against a rich equivariant family pins the procedure to the equivariant class): Eaton, *Group Invariance Applications in Statistics* (1989); Lehmann & Romano, *Testing Statistical Hypotheses* (invariance chapter); Berger, *Statistical Decision Theory*. This is the correct, \$50-year-old provenance of our Theorem A forward fact and Lemma 2’s converse.
- **Structure of equivariant maps / universal approximation** — Maron et al., *On the Universality of Invariant Networks* (2019); Yarotsky, *Universal approximations of invariant maps* (2022); Dym & Maron, *Universality of Rotation-Equivariant Point Cloud Networks* (2021); Goodman & Wallach, *Symmetry, Representations, and Invariants* (the Sym^k / Λ^k decompositions behind Proposition 4).
- **Predictability horizon / perturbation growth / ergodic rate** — Lorenz, *Atmospheric predictability* (1969); Oseledets, *A multiplicative ergodic theorem* (1968) — the local/finite-time exponent converges a.e. to the asymptotic rate λ_1 , the rigorous half of Proposition 7; Pilyugin, *Shadowing in Dynamical Systems* (1999) — on *uniformly* hyperbolic sets, bounds a perturbed model’s forecast-horizon *floor* (orbit error, not the exponent); Tucker, *A rigorous ODE solver and Smale’s 14th problem* (2002) and Araújo–Pacífico–Pujals–Viana (2009) — the geometric Lorenz attractor is *singular*-hyperbolic with an SRB measure (so the MET applies) but is **not** uniformly hyperbolic (so classical shadowing does not formally cover it — hence E2’s exponent-lift is empirical). The $T(\epsilon) \sim \log(1/\epsilon)/\lambda$ law and the $e^{\lambda T}$ growth of a seeded perturbation are classical; our twist is to seed the growth with the *equivariance residual* (Proposition 6) and to characterize *when* the local spectrum governs a learned model’s horizon (Proposition 7).

Companion line. This paper is a focused extraction of the project’s *Certified World Models* program; the full experiment suite, the closed-loop clause, the discovery/generation re-framing, and the complete proofs appear in the companion manuscript paper2_certified_world_models.md.

A. ADDITIONAL FIGURES

Supporting figures relocated from the main text (the main-text figures are the hero schematic, the tightness construction, the Lorenz horizon lift, and the real-contact-dynamics certificate).

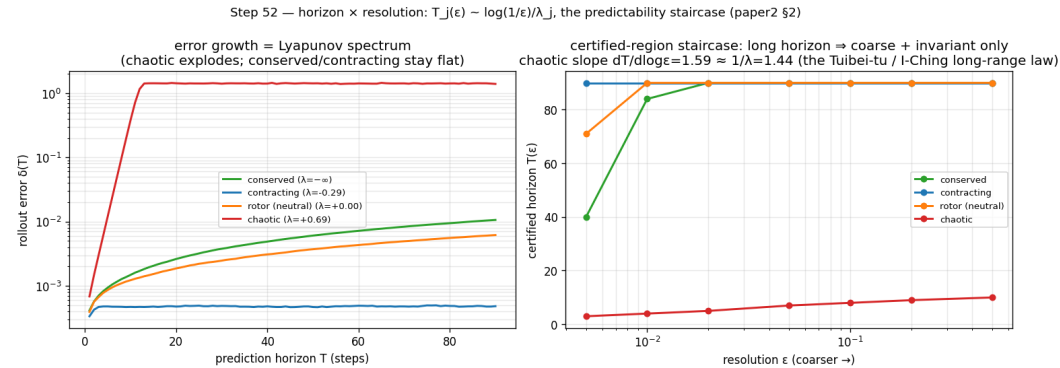


Figure 5: Figure A1. The horizon $\times$ resolution staircase on a controlled spectrum (E2). The measured certified horizon per channel as the demanded resolution ϵ sharpens, recovering $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (slope 1.3–1.6): chaotic channels are certified to a few steps, slow/invariant channels to dozens.

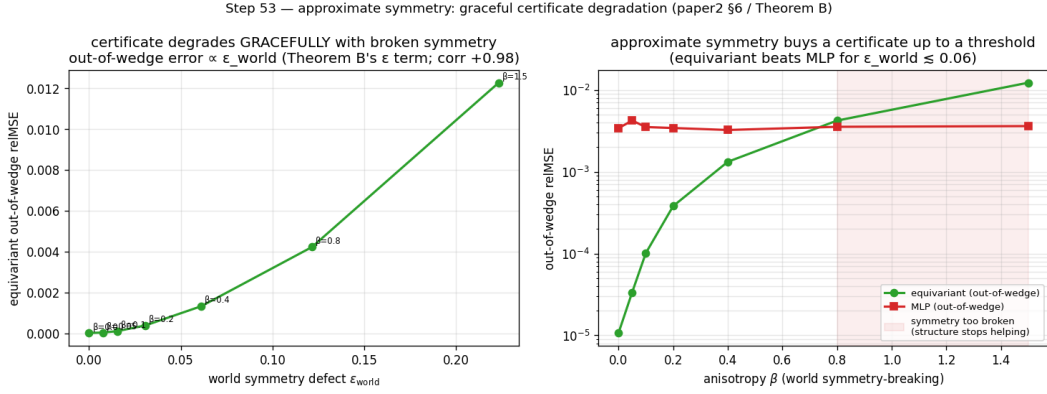


Figure 6: Figure A2. Approximate symmetry (E3). **Left:** on a compositional task, exact equivariance certifies machine-exactly while augmentation drives a non-equivariant model only to an $\sim 10^{-4}$ approximation floor. **Right:** breaking the world’s symmetry by ϵ_{world} degrades the certificate gracefully and *linearly* (the slope Proposition 6 predicts) up to a measured threshold.

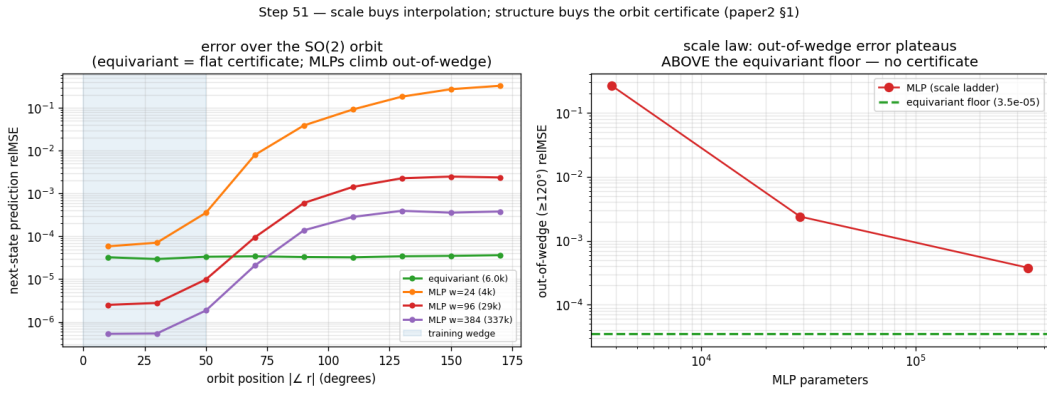


Figure 7: Figure A3. Structure versus scale (E4). The equivariant model is orbit-flat (a certificate); an $88\times$ -scaled non-equivariant model buys *in-distribution* interpolation (it beats the equivariant model inside the training wedge) but its error climbs $10\text{--}155\times$ above the equivariant floor *out* of the wedge — scale buys interpolation, not the certificate.

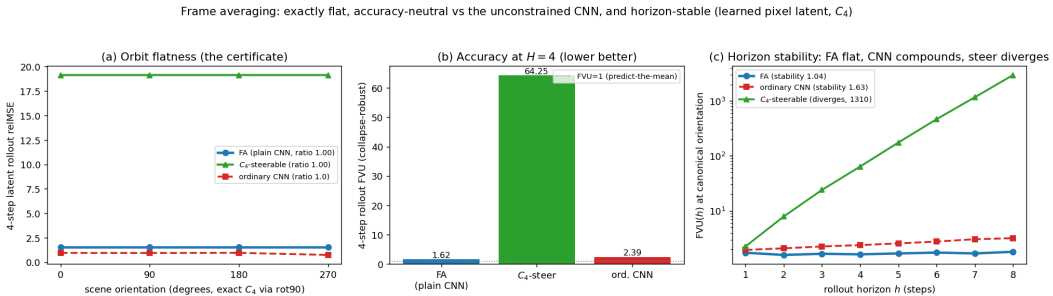


Figure 8: Figure A4. The certificate on raw pixels at no accuracy cost (E6). On rendered PushT frames (exact C_4), frame averaging makes a plain CNN/MLP exactly C_4 -equivariant: the latent rollout is flat over the orbit (left, the certificate), it matches or beats an unconstrained CNN on a collapse-robust accuracy metric (centre), and its rollout error stays bounded over horizon while the steerable baseline diverges (right, log scale).

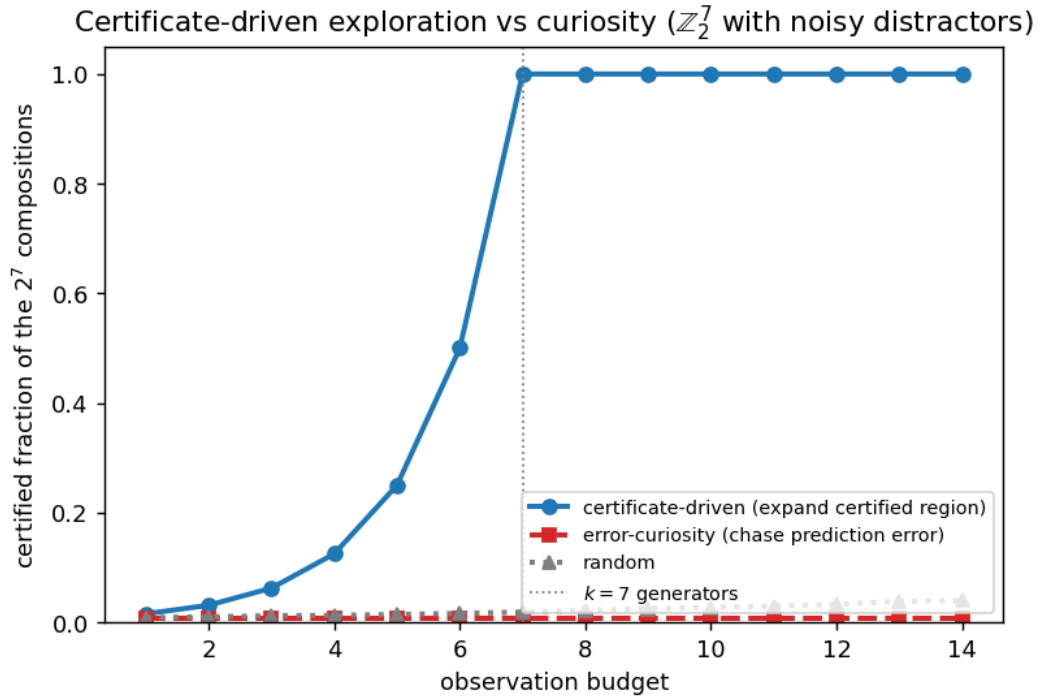


Figure 9: Figure A5. Certificate-driven active inference (E8). On a $\mathbb{Z}_2^7$ compositional world with noisy distractor actions, an explorer that maximizes the *certified region* certifies all 128 compositions in exactly 7 observations (the generator basis), while a prediction-error-curiosity explorer is lured by the high-error distractors and certifies almost nothing at the same budget.